Variable-Delay Low-Pass, All-Pass Filters and their Applications

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Department of Electrical, Electronics and Communication Engineering

॥ सा विद्या या विमुक्तये ॥
ಭಾ.ತಂ.ಸಂ. ಧಾರವಾಡ
भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

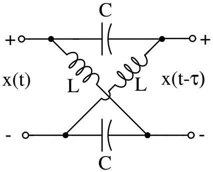
10^{th} May 2025

Overview

Overview

Introduction to Variable-Delay Filters

All-Pass Filter



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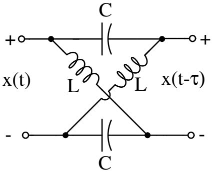

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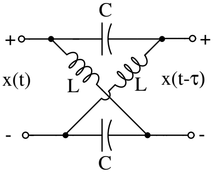

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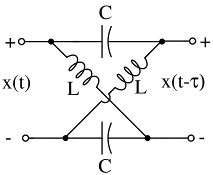

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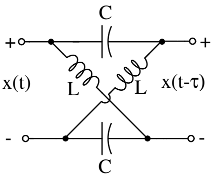

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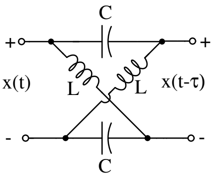
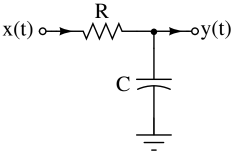


Low-Pass Filter



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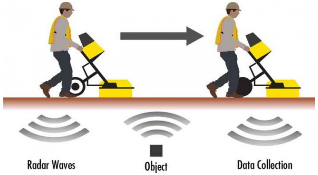
- Delay of low-pass filters (LPF) by changing time constant (RC) of filter

Applications of Variable-Delay Filters

Expansion of narrow sporadic pulses

Expansion of narrow sporadic pulses



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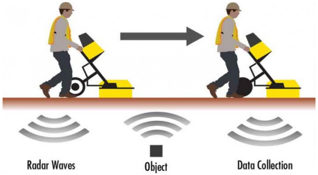

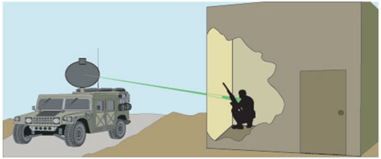

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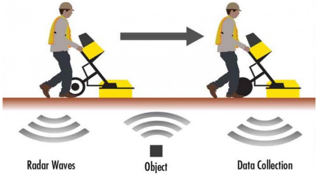

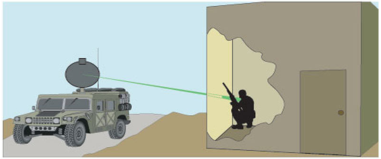

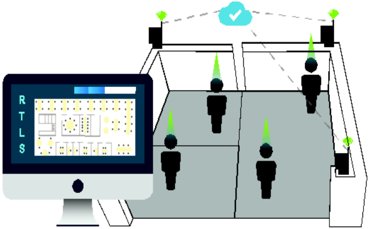

Image courtesy: https://scantech.ie/scantech-about-gpr.html

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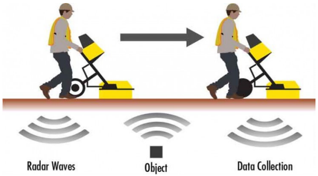

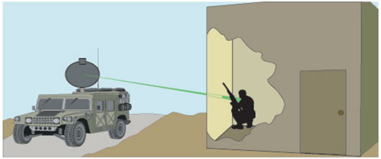

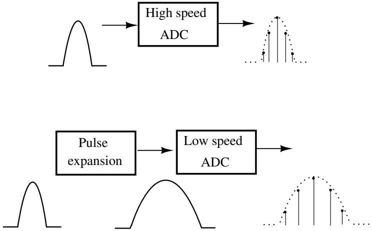

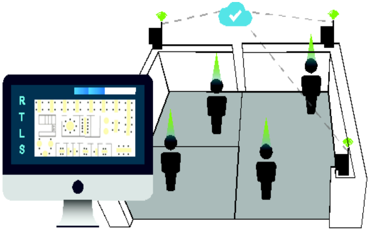

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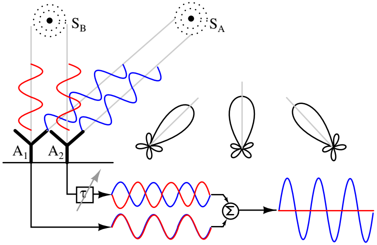
TTD based beamforming



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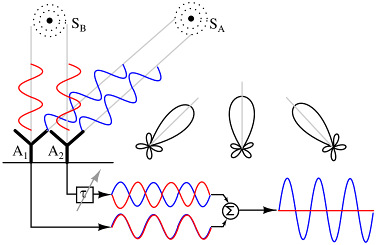
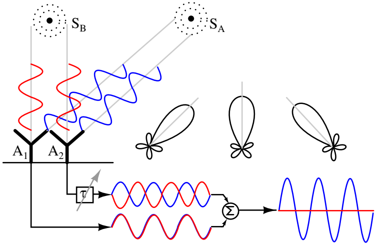


- Variable-delay APF in equalizers

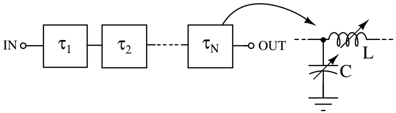
TTD based beamforming



- Variable-delay APF in equalizers
- Variable-delay LPF in voltage-controlled oscillators

Overview

Variable-delay transmission line[1]



[1] Chu et al., IEEE JSSC, 2007.

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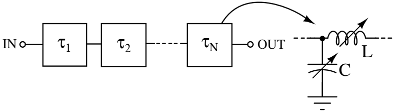

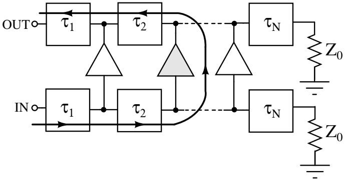

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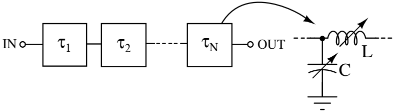
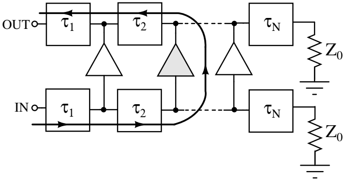
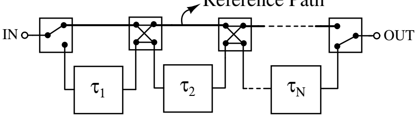




Variable-delay Passive APF[2]



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Reported Variable-Delay Filters

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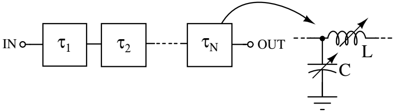

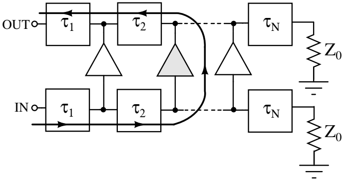

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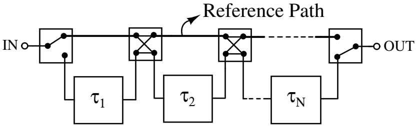

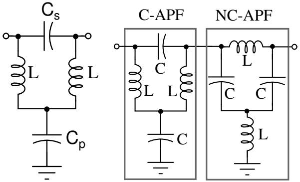

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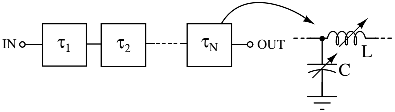

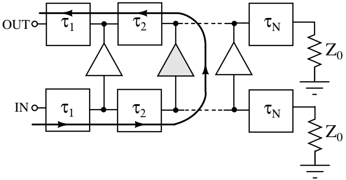

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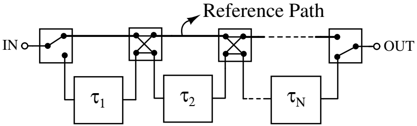

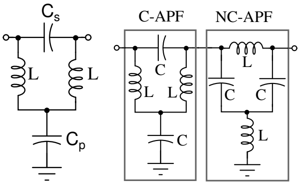

- Programmable delay lines in beamforming applications

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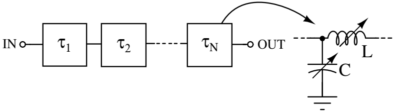

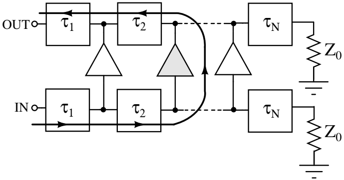

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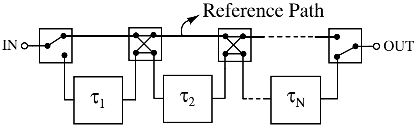

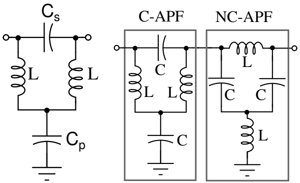

- Programmable delay lines in beamforming applications
- Suffer from low Q of L,C, and other switch losses

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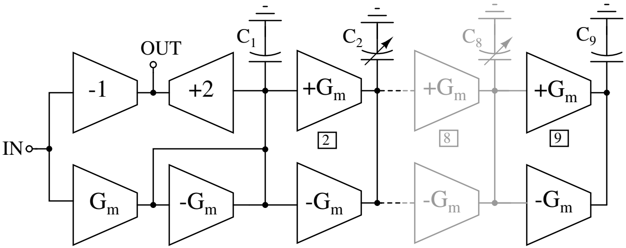
Reported Variable-Delay Filters

Variable order constant bandwidth filter[3]



- Delay tuning by changing the order of the filter

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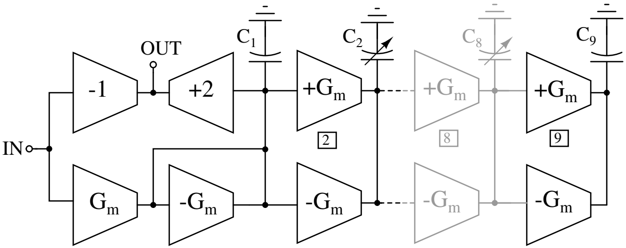

- Delay tuning by changing the order of the filter
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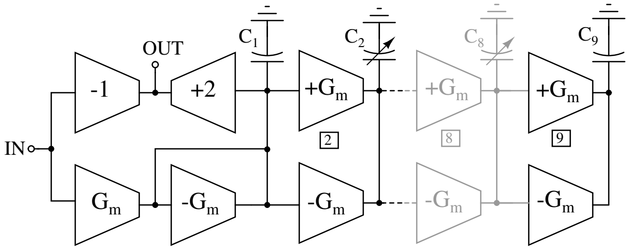

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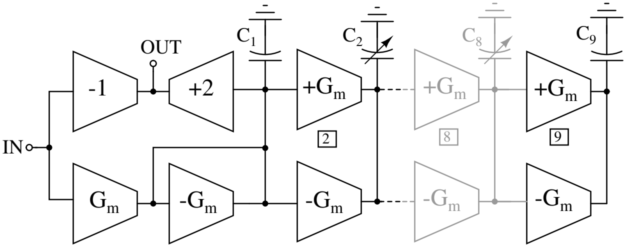

- Delay tuning by changing the order of the filter
- Constant bandwidth with varactors
- Used as wideband TTD circuit
- Variable order APF requires a large range of C and hence increased power and occupies a large area

[3] Mondal et al., IEEE JSSC, 2017

Reported Variable-Delay Filters

Variable bandwidth filter: tunable-C [4],[5]

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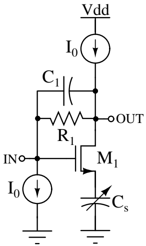

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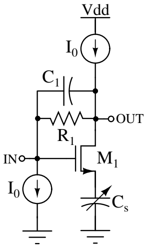
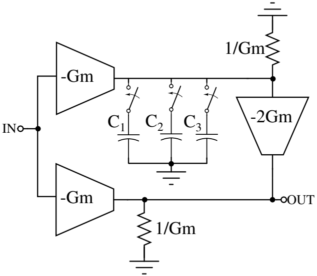
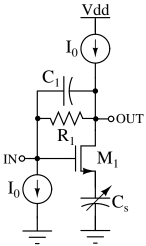
Variable bandwidth filter: tunable-C [4],[5]

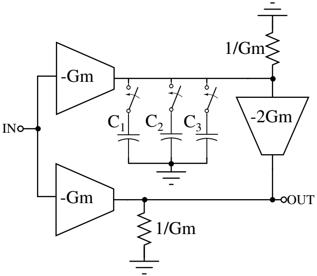




[4] Elamien et al. IEEE MWCL 2021
[5] Garakoui et al., IEEE JSSC, 2015.

Variable bandwidth filter: tunable-C [4],[5]





- Variable C active Gm-C in programmable delay lines

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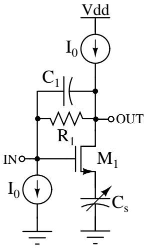
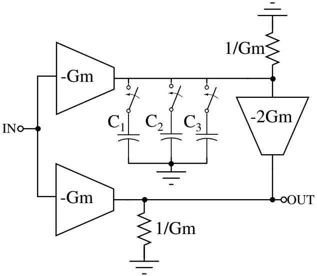
Variable bandwidth filter: tunable-C [4],[5]





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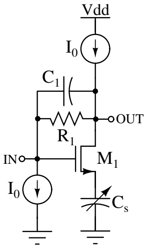
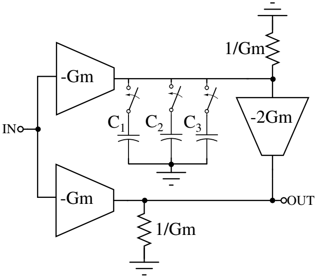
Variable bandwidth filter: tunable-C [4],[5]

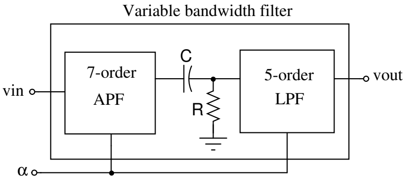




- Variable C active Gm-C in programmable delay lines
- Low Q-varactors and requires additional buffer stages
- 1-order Gm-C active APF with switched C requires an additional gain enhancement circuit

[4] Elamien et al. IEEE MWCL 2021
[5] Garakoui et al., IEEE JSSC, 2015.

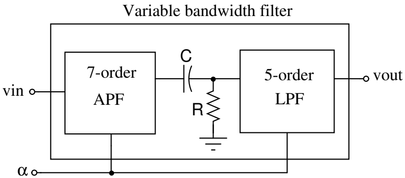
Variable bandwidth filter: tunable-Gm [6]



[6] Mondal et al., IEEE TCAS1, 2017.

Reported Variable-Delay Filters

Variable bandwidth filter: tunable-Gm [6]



- The reported technique used in expansion and compression techniques

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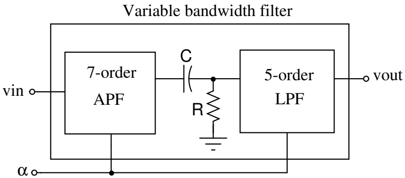

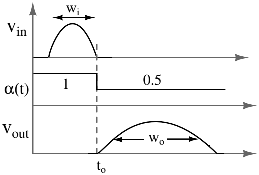

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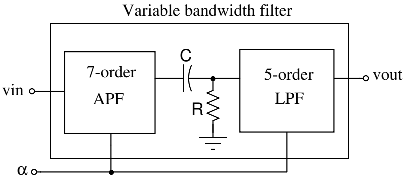
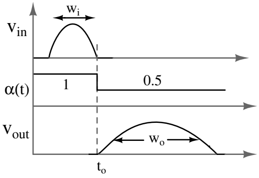
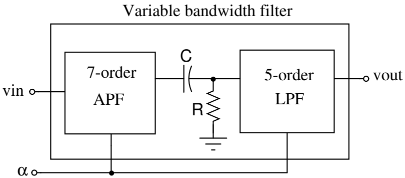


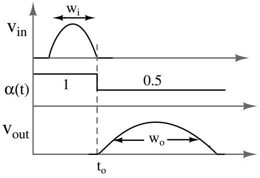


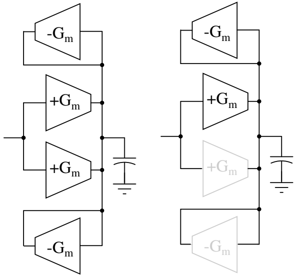
- The reported technique used in expansion and compression techniques
- Increase the delay of the filter by reducing the bandwidth

[6] Mondal et al., IEEE TCAS1, 2017.

Variable bandwidth filter: tunable-Gm [6]







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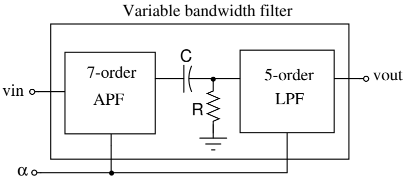
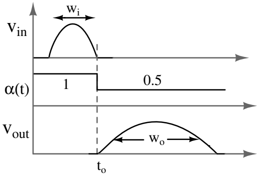
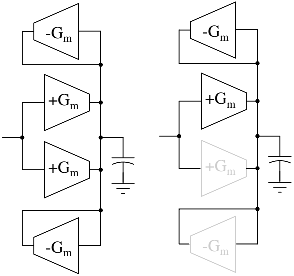
- Increase the delay of the filter by reducing the bandwidth

[6] Mondal et al., IEEE TCAS1, 2017.

Variable bandwidth filter: tunable-Gm [6]







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Reported Variable-Delay Filters

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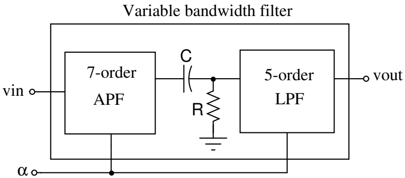
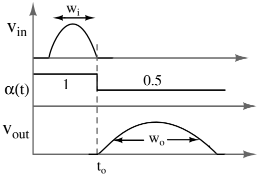
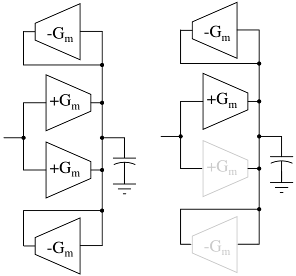


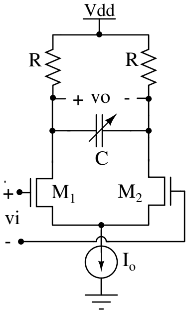




- The reported technique used in expansion and compression techniques
- Increase the delay of the filter by reducing the bandwidth
- Uses constant-C frequency scaling with programmable Gms
- Higher order filters and Gms to be switched instantaneously and precisely

[6] Mondal et al., IEEE TCAS1, 2017.

Variable-delay LPF[7]



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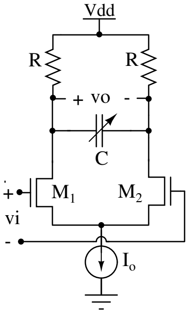
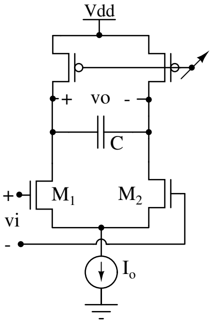


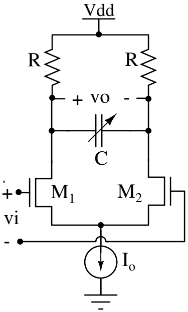


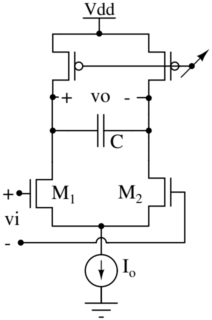
- Varactors or switched C to change delay of LPF
- Variable R from voltage-controlled PMOS

[7] X.Gui et al., IEEE TCAS1, 2013.

Reported Variable-Delay Filters

Variable-delay LPF[7]





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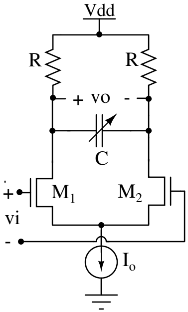

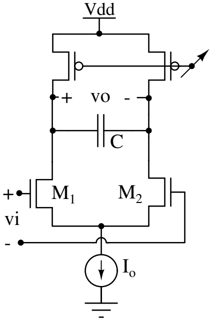

- Varactors or switched C to change delay of LPF
- Variable R from voltage-controlled PMOS
- Variable-delay LPF in ring-VCO
- Require high Q-C with constant oscillation amplitudes over tuning range of VCO

[7] X.Gui et al., IEEE TCAS1, 2013.

Overview

1. Introduction to Variable-Delay Filters and their Applications
2. Literature Survey on Variable-Delay Filters
3. Thesis Contribution
4. Variable-Delay LPF using Miller Effect
   - TTD element
   - Pulse Expansion and Compression
   - 3-Stage Ring VCO
5. Variable-Delay Gm-C APF with tuned Gm
   - TTD element
   - Pulse Expansion and Compression
   - Inline Calibration for High Gain OTA
6. Study of TTD non-idealities in Timed Array Receivers
7. Conclusion and Future Work

Thesis Contribution

- **Contribution-1**
  - TTD circuit from variable-delay LPF with the principle of Miller effect
  - Applied to pulse expansion, compression, and ring-VCO circuits

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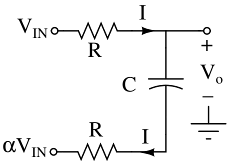
- **Contribution-3**
  - Study of TTD non-idealities on timed array receivers

Overview

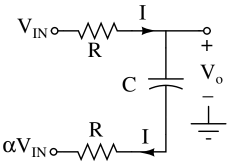
Variable-Delay LPF(*)



- Tuning of impedance (V_{IN}/I) and hence delay from α using Miller effect

* Naveen K, Mayur S,"A Circuit for Expanding, Compressing, or Delaying a Signal"-Indian Patent 555660"

Variable-Delay LPF$^{(*)}$



$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

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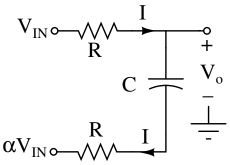

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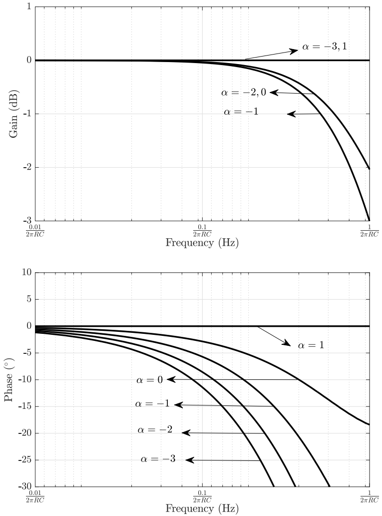

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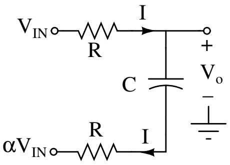

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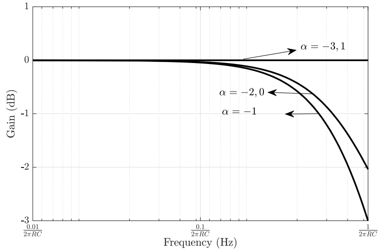
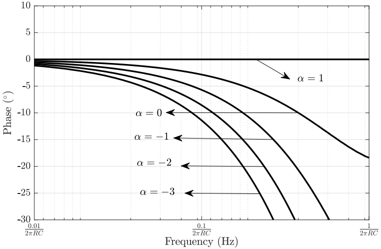
- Tuning of impedance (V_{IN}/I) and hence delay from α using Miller effect
- Delay tuning techniques in TTD circuits, pulse expansion (compression), and VCO applications





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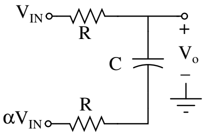
Overview

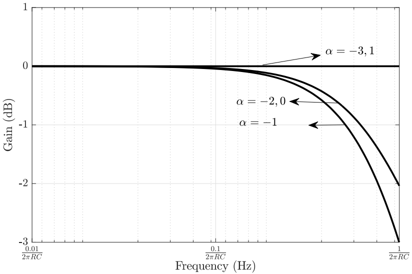
Variable-Delay LPF as TTD Circuit

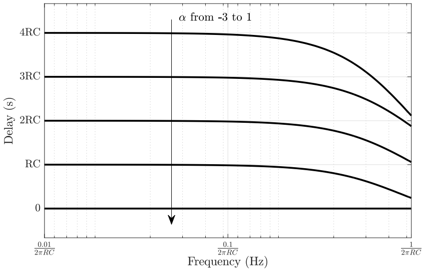


$$H_\alpha(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

$$\tau_g = -d(\measuredangle H(j\omega))/d\omega$$
$$\approx (1-\alpha)RC$$

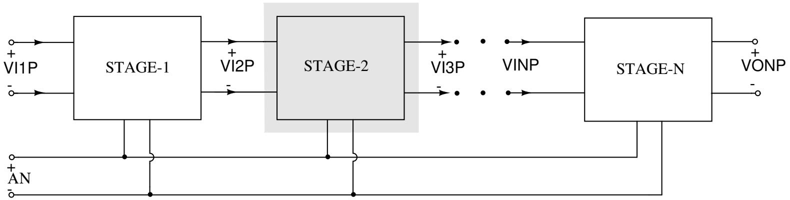
- $H_\alpha(s)$ is LPF for $-3 < \alpha < 1$
- $|H_\alpha(s)| = 1$ for $\alpha = -3, 1$





Variable-Delay LPF as TTD Circuit

- Identical stages are cascaded to increase delay range



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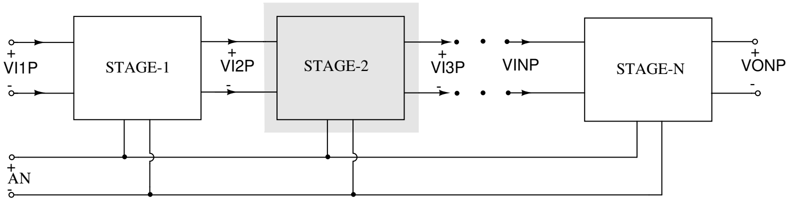
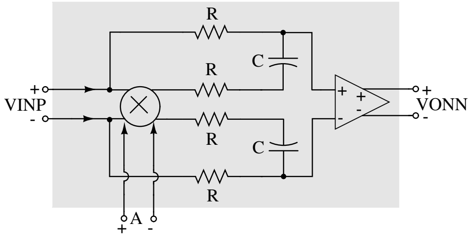
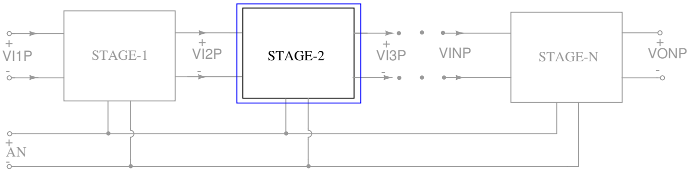


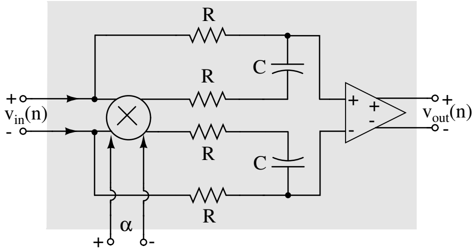
Differential architecture

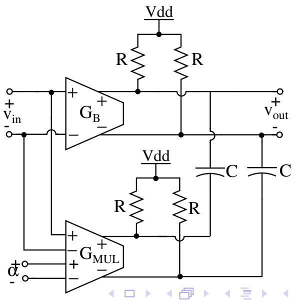


TTD Circuit with Cascaded LPFs

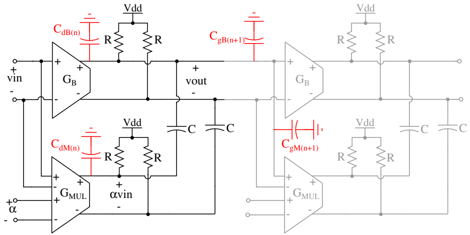


Circuit Level Implementation





Effect of Parasitics on TTD



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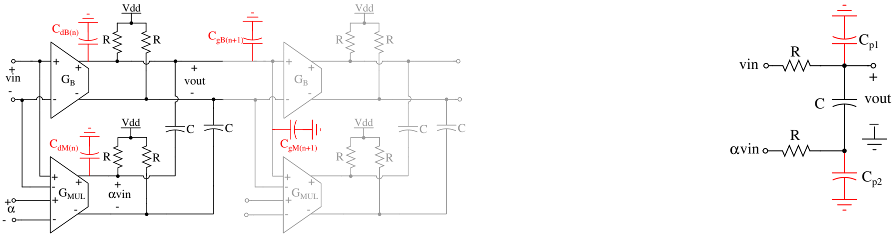

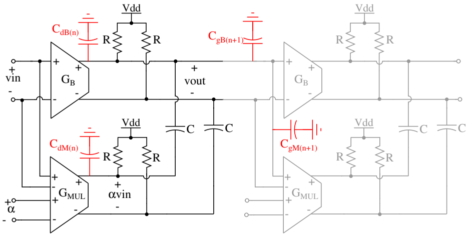

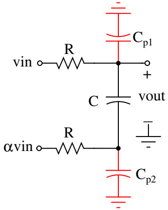

With: $C_{p1} = xC$, $C_{p2} = yC$.

$$H_\alpha(s) \approx \frac{(1 + sRC[(1+\alpha+y])}{\Big(1 + sRC[2 + (x+y)/2]\Big)\Big(1 + sRC[(x+y)/2]\Big)}$$

$$\tau_\alpha \approx \big[(1 + x - \alpha)\big] RC$$

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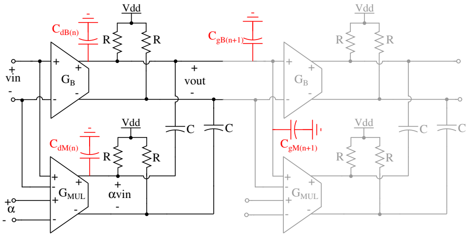

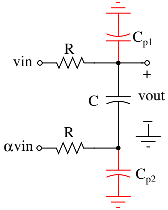

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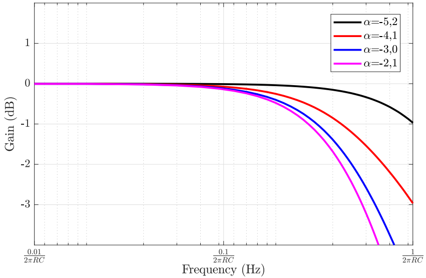
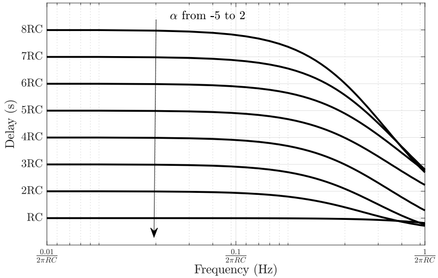
- $-(3+x) \leq \alpha < (1+x)$.

Effect of parasitics on TTD

Actual implementation shows $C_{p1} \approx 2C$, $C_{p2} \approx 0.5C$





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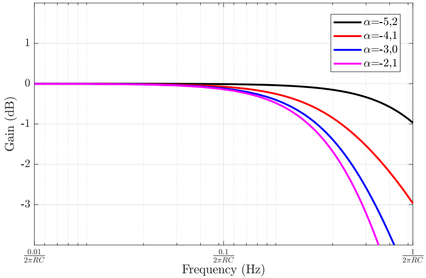
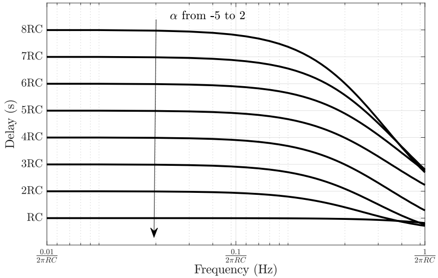


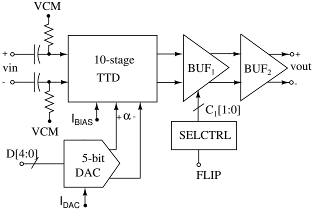


variable-delay LPF	without parasitics	with parasitics
α	$-3 \leq \alpha \leq 1$	-5 $\leq \alpha < 3$
Delay range	4RC	7RC
Bandwidth	0.1/(2πRC)	0.07/(2πRC)

- Effect on Delay range-Bandwidth factor of TTD from N-stage LPFs

10-Stage TTD Circuit: Post-Layout Simulation Results

Designed in UMC-180 nm technology



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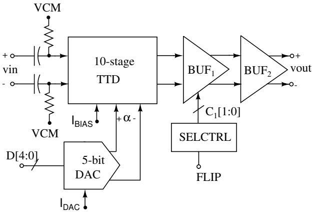

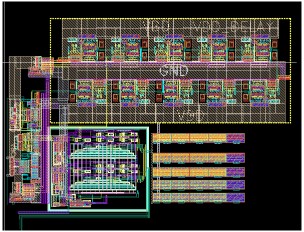

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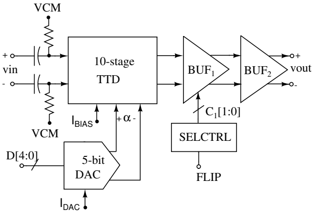

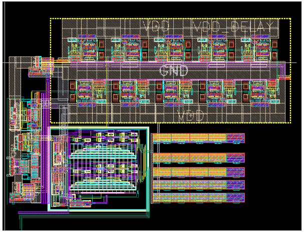

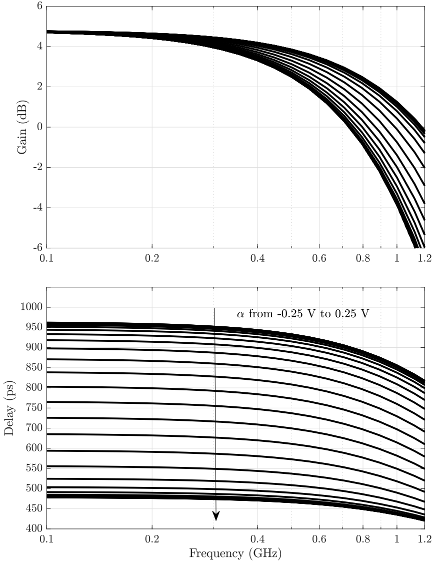

Need for Delay Correction

$$\tau_g = -d(\measuredangle H(j\omega))/dw \approx (1-\alpha)RC$$

- Delays are not constant across all the frequencies

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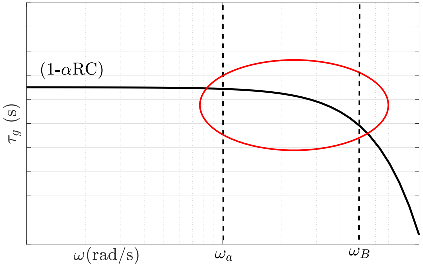

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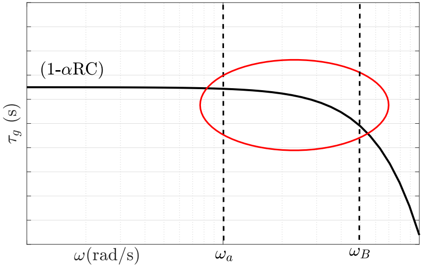

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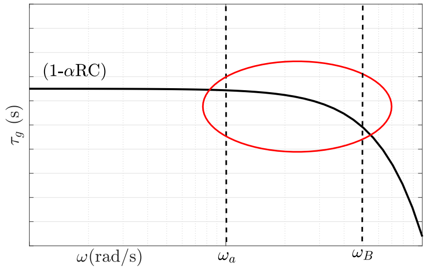
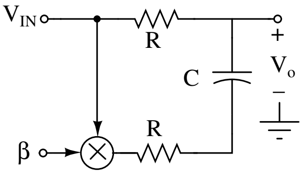
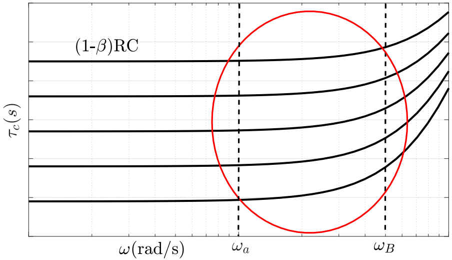


- Delay remains constant upto ω_a and then decreases
- Delay correction technique to keep the delay constant up to the required frequencies

Circuit for generating Delays with positive slope $d\tau_\beta/d\omega$

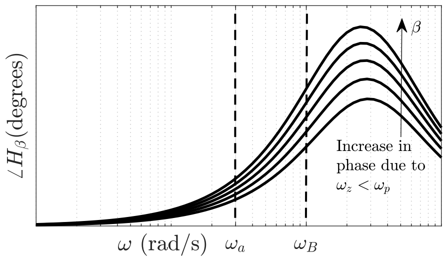
- Delay with positive slope is obtained from a similar 1-order RC circuit



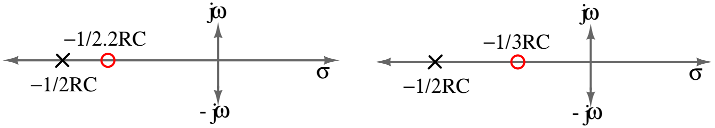


- Control signal is varied with positive value of $1.2 \leq \beta \leq 2$

Circuit for generating Delays with positive slope $d\tau_\beta/d\omega$

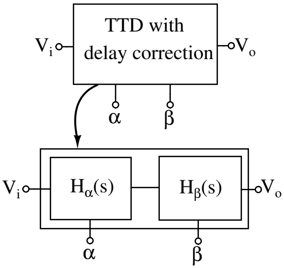


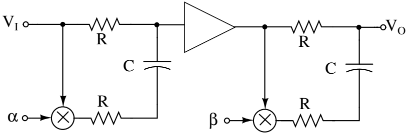
- Increase in phase response for increase in β.



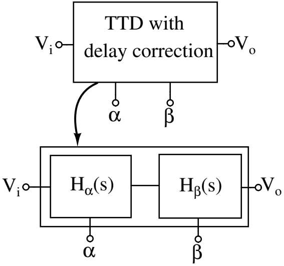
- Occurrence of LH-zero before pole

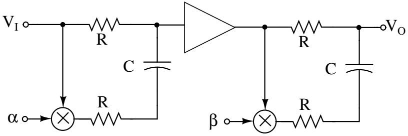
TTD with Delay Correction





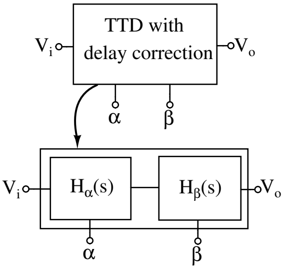
TTD with Delay Correction

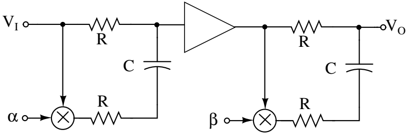




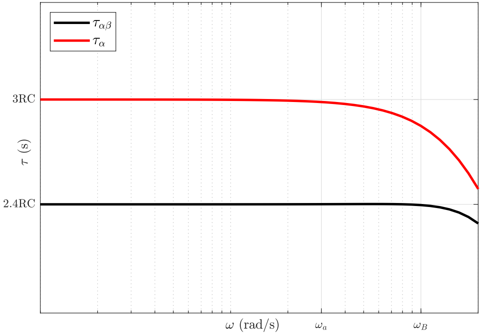
- β is determined by inspection for $d\tau_\alpha/d\omega + d\tau_\beta/d\omega = 0$

TTD with Delay Correction

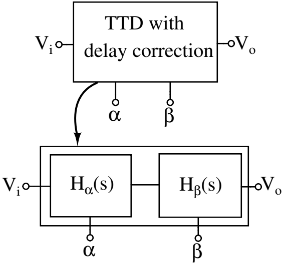


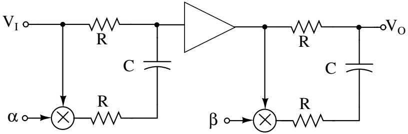


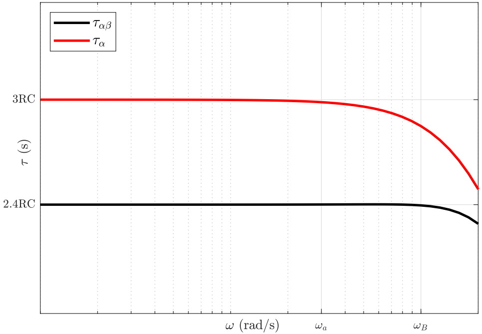
- β is determined by inspection for $d\tau_\alpha/d\omega + d\tau_\beta/d\omega = 0$
- $\alpha = -2$, $\beta = 1.6$ constant delay up to ω_B



TTD with Delay Correction

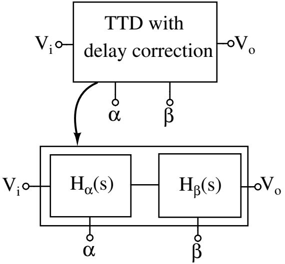


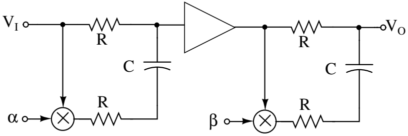


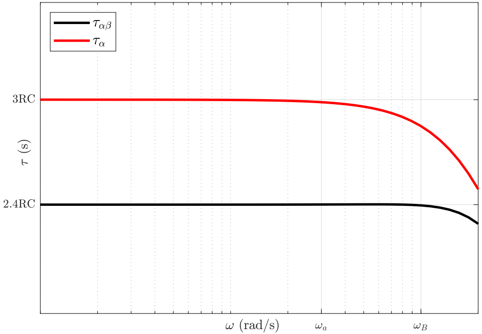


- β is determined by inspection for $d\tau_\alpha/d\omega + d\tau_\beta/d\omega = 0$
  - $\alpha = -2$, $\beta = 1.6$ constant delay up to ω_B
  - improve in bandwidth by factor of 2.5

TTD with Delay Correction



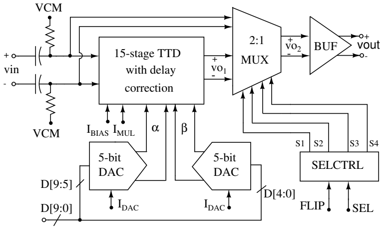




- β is determined by inspection for $d\tau_\alpha/d\omega + d\tau_\beta/d\omega = 0$
  - $\alpha = -2$, $\beta = 1.6$ constant delay up to ω_B
  - improve in bandwidth by factor of 2.5
  - increase in power by factor of 2

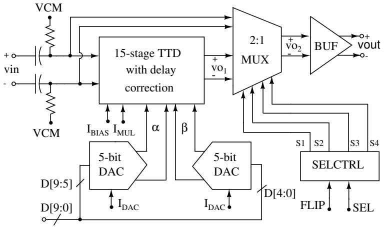
15-Stage TTD Circuit: Post-Layout Simulation Results

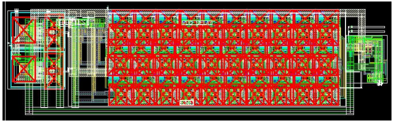
Designed in TSMC 65-nm technology



15-Stage TTD Circuit: Post-Layout Simulation Results

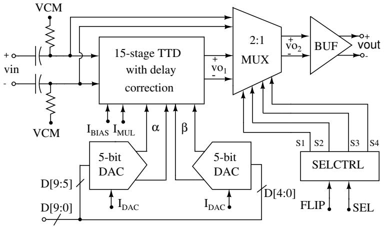
Designed in TSMC 65-nm technology

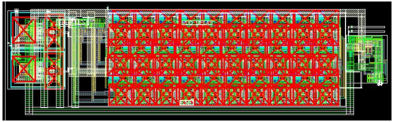


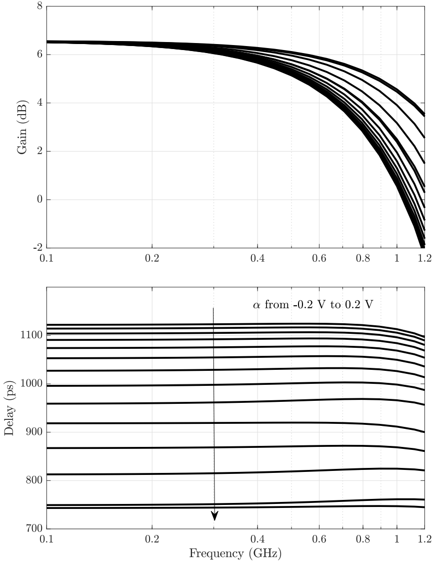


15-Stage TTD Circuit: Post-Layout Simulation Results

Designed in TSMC 65-nm technology







Results and Comparisons

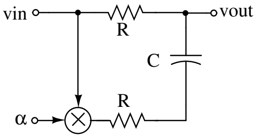
Parameters	This work	This work	This work	[MTTS-2015]	[JSSC-2015]	[JSSC-2017]	[MWCL-2021]
Technology (nm)	65	65	180	130	140	130	65
Architecture	LPF-with correc.	LPF-without correc.	LPF	P-APF	A-APF	A-APF	A-APF
Maximum delay(ps)	1150	1000	955	400	550	1700	55
Bandwidth (GHz)	DC-1	DC-1	DC-1	1-20	1-2.5	DC-2	0.4-2
Delay Range (ps)	400	420	450	350	550	1450	27
Delay Variation (ps)	±5	±50	±60	±8	±10	±140	±2
Gain Variation (dB)	±2.5	±6	±4	±15	±1.4	±0.7	±0.375
Power (mW)	36	18	16	6	450	364	2.47
FOM-1	-20.92	-27.35	-28.98	-60.89	-5.26	-18.02	-24.85
FOM-2	-47.95	-27.82	-32.46	-57.69	-31.25	-30.91	-40.69

FOM-1=20log(Power/(delay range $\times$ bandwidth))

FOM-2=FOM-1+20log($\frac{2\times\text{delay variation}}{\text{delay range}}$)+2$\times$gain variation

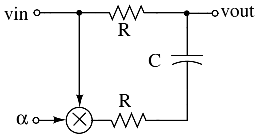
Overview

Variable-delay LPF as Expansion (Compression) Circuit

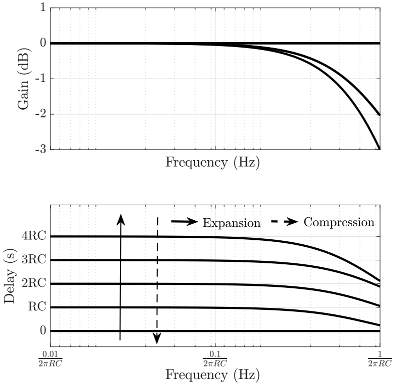


$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

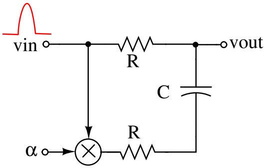
Variable-delay LPF as Expansion (Compression) Circuit



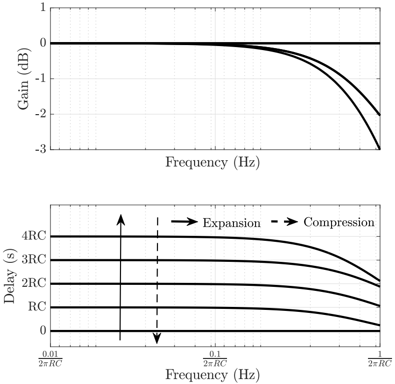
$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$



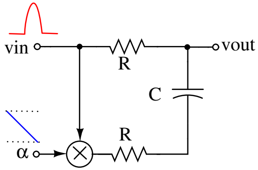
Variable-Delay LPF as Expansion (Compression) Circuit



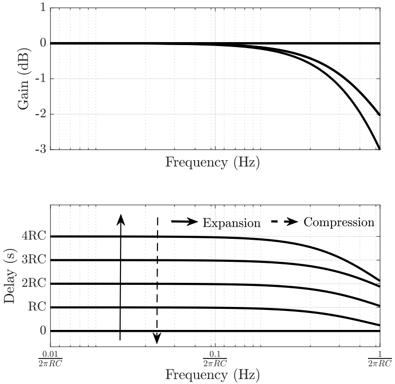
$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$



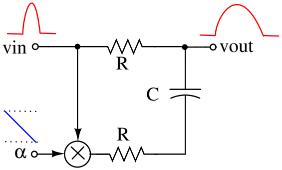
Variable-Delay LPF as Expansion (Compression) Circuit



$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

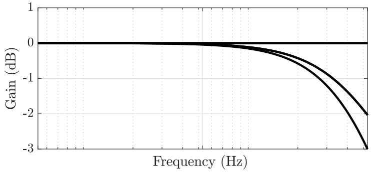


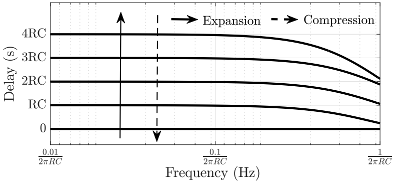
Variable-Delay LPF as Expansion (Compression) Circuit



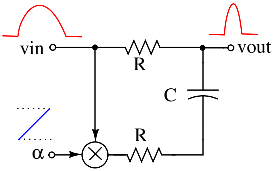
$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

- α to be decreased linearly for expansion



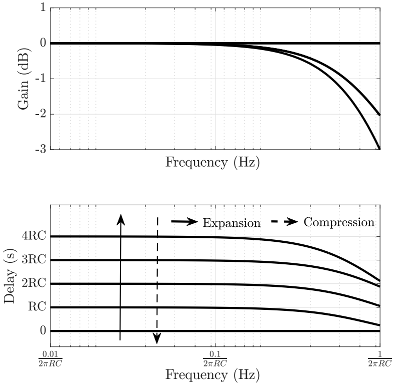


Variable-Delay LPF as Expansion (Compression) Circuit

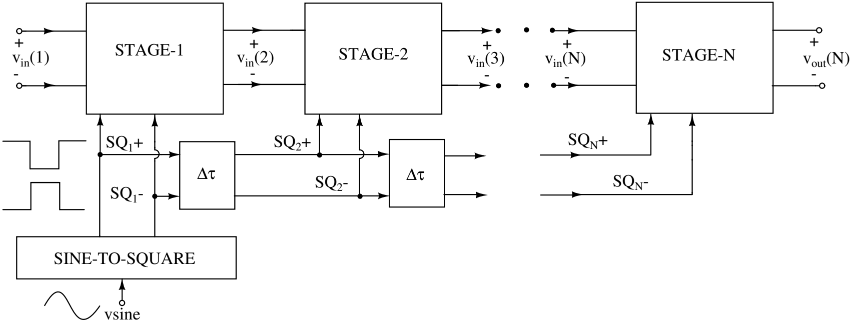


$$H(s) = \frac{1 + (1+\alpha)sRC}{(1 + s2RC)}$$

- α to be increased linearly for compression

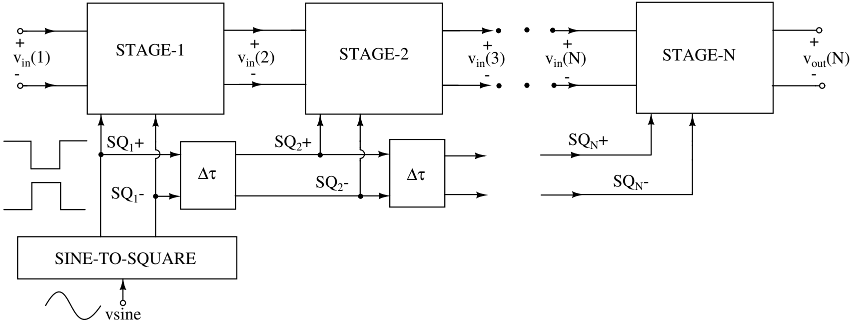


N-stage Expansion (Compression) Circuit



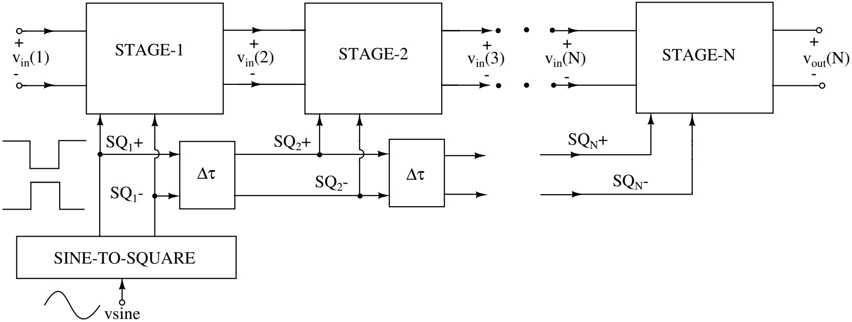
- Multiple expansion(compression) stages are cascaded

N-stage Expansion (Compression) Circuit



- Multiple expansion(compression) stages are cascaded
- Differential ramp signal for expansion(compression) is obtained from SINE-TO-SQUARE block

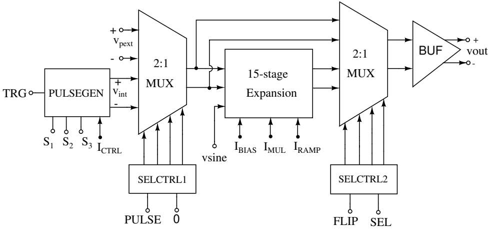
N-stage Expansion (Compression) Circuit



- Multiple expansion(compression) stages are cascaded
- Differential ramp signal for expansion(compression) is obtained from SINE-TO-SQUARE block
- The control signals are delayed for succeeding expansion(compression) stage

15-Stage Expansion Circuit: Post-Layout Simulation Results

Designed in UMC 180-nm and TSMC 65-nm technologies



15-Stage Expansion Circuit: Post-Layout Simulation Results

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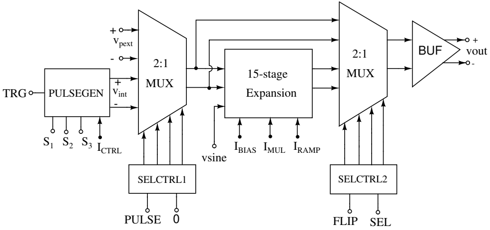

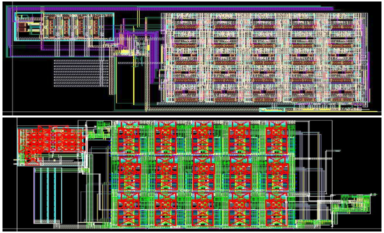

15-Stage Expansion Circuit: Post-Layout Simulation Results

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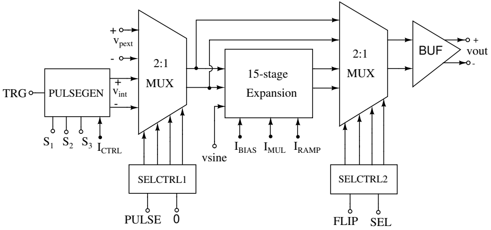
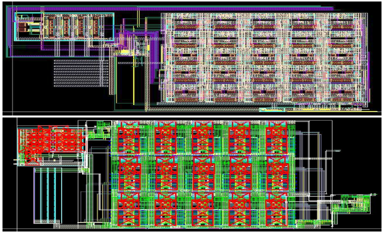
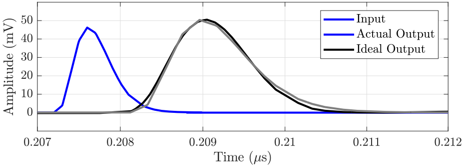
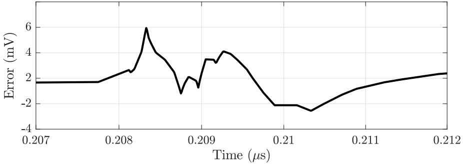




TSMC 65-nm results



(a)

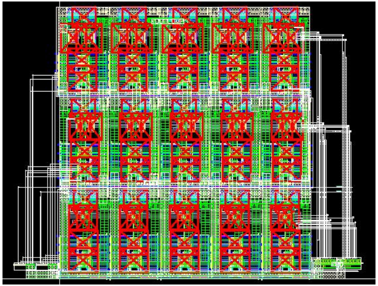


(b)

- Input pulse: half power pulse width 350 ps
- Stretch factor 2.25, RMSE=1.5 mV

15-Stage Compression Circuit: Post-Layout Simulation Results

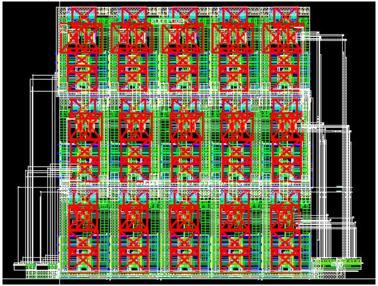
Designed in TSMC 65-nm technologies



- Ideal Gaussian input pulse: 3.32 ns
- Compression factor 0.47
- RMSE 1.05mV

15-Stage Compression Circuit: Post-Layout Simulation Results

Designed in TSMC 65-nm technologies



- Ideal Gaussian input pulse: 3.32 ns
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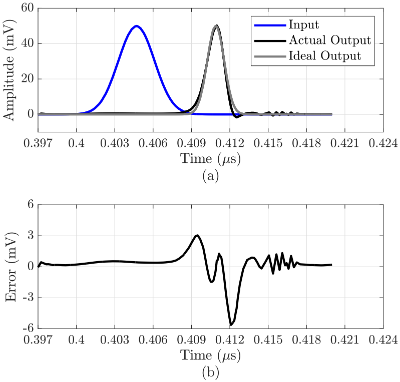


(a)

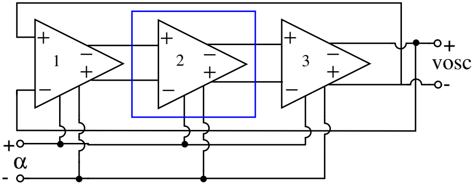
(b)

Comparison with Reported Expansion Techniques

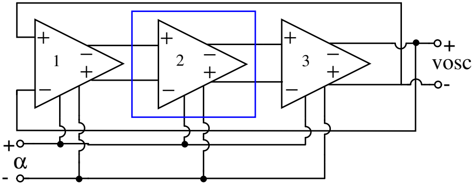
	This work	This work	[TCAS1-2018]	[ISSCC-2015]	[ICRF-2012]	[TCAS1-2012]
Techniques	Variable delay LPF	Variable delay LPF	Variable bandwidth filter	Sampling	Dispersive filter	Photonic
Technology (nm)	65	180	130	65	130	Discrete
Carrier/clock	Not required	Not required	Not required	Sampling clock	Chirp carrier from VCO	Optical chirp carrier
Bandwidth (GHz)	1	1	0.87	3.75-4.25	1	5
Stretch factor	2.25	2.2	1.8	400	2	10
Power(mW)	36	110	350.61	170	-	-

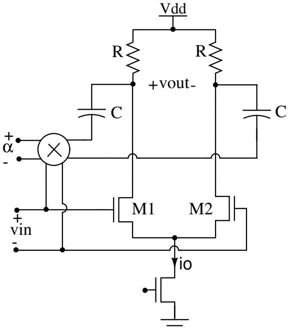
Overview

3-Stage VCO with Proposed Delay Cell

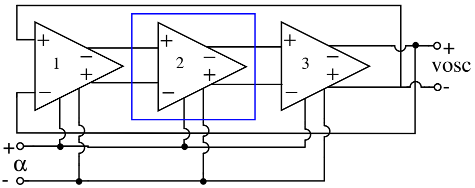


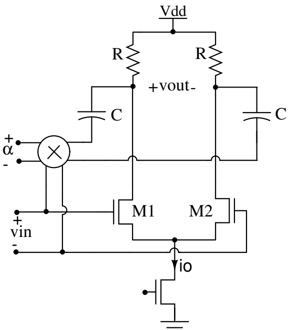
3-Stage VCO with Proposed Delay Cell





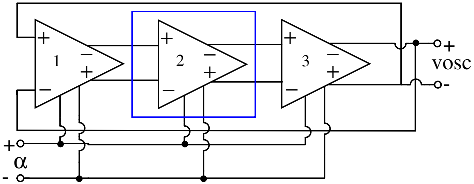
3-Stage VCO with Proposed Delay Cell

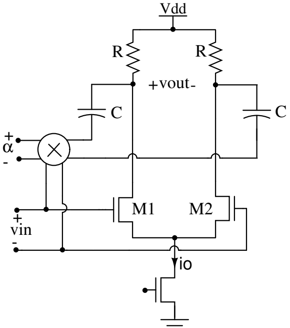




- Change of C with Miller effect

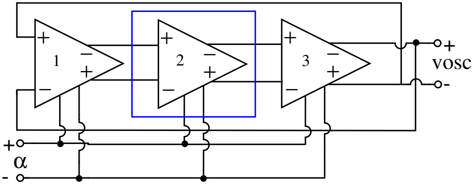
3-Stage VCO with Proposed Delay Cell

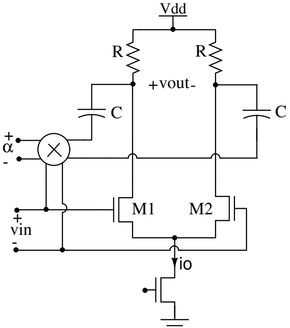




- Change of C with Miller effect
- Tuned oscillations with α

3-Stage VCO with Proposed Delay Cell

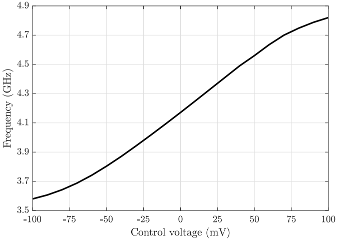


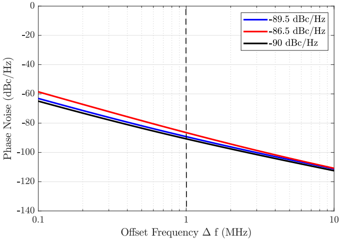


- Change of C with Miller effect
- Tuned oscillations with α
- High Q varactors are not required and maintain constant oscillation amplitude

Simulation Results

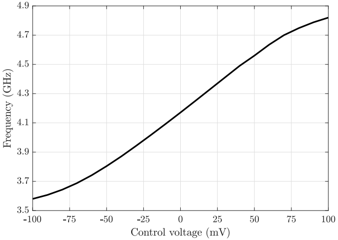
TSMC 180-nm

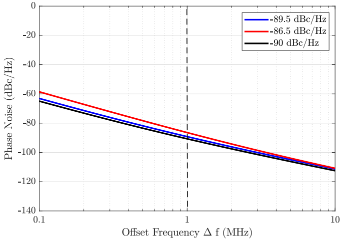




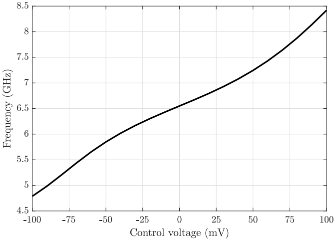
Simulation Results

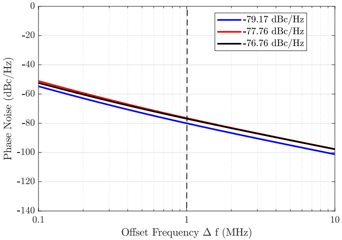
TSMC 180-nm





TSMC 65-nm





Comparison with reported CML-VCOs

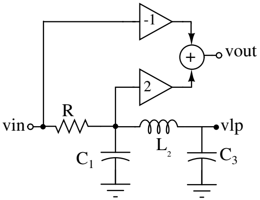
	This work	This work	[TCAS1-2013]	[MWCL-2020]
Technology (nm)	180	65	180	180
Frequency range (GHz)	3.58-4.8	4.7-8.4	1.78-2.53	2.2-2.7
Supply voltage (V)	1.8	1.2	1.8	1.8
Phase noise (dBc/Hz @1MHz)	-86.5	-76.76	-92.68	-94.41
Power (mW)	13.3	7.2	26	10.1
FOM* (dBc/Hz)	177.41	181.69	159.19	176.04

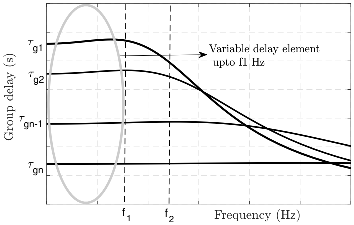
$$* FOM = 20\log[f_o/\Delta f] - L(\Delta f) - 10\log(P_{DC}) + 20\log[FTR/10]$$

- Higher FOM compared to other CML delay cell-based ring VCOs.

Overview

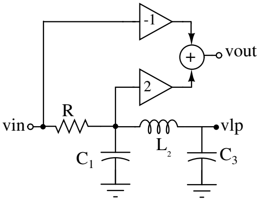
Variable-Delay APF*

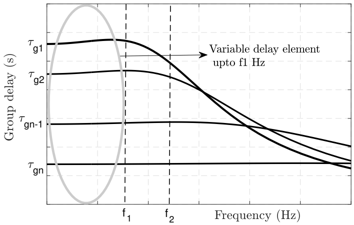




* Mayur S.M. et al. IEEE ISCAS 2022

Variable-Delay APF*





- Changing L,C values varies delay τ_g

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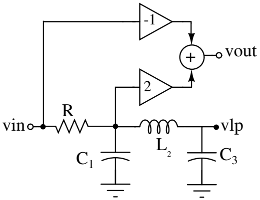

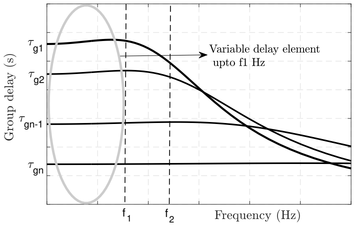

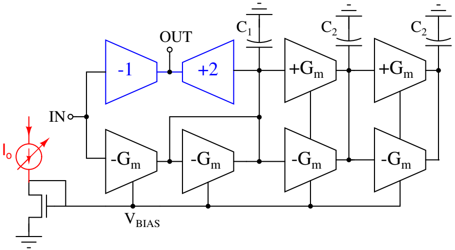

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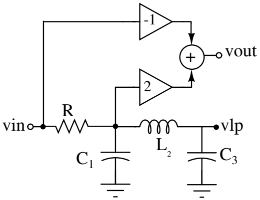

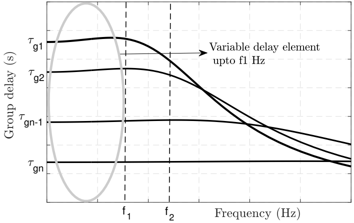

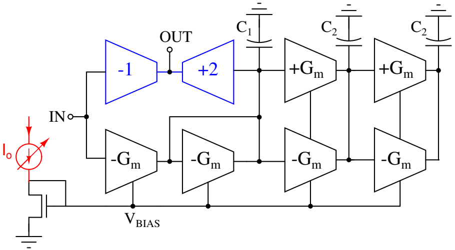

- Changing L,C values varies delay τ_g
- In Gm-C implementation of APF
  $\tau_g \approx (\frac{2}{G_m})(C_1 + C_3)$

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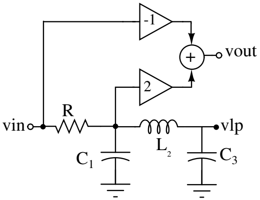
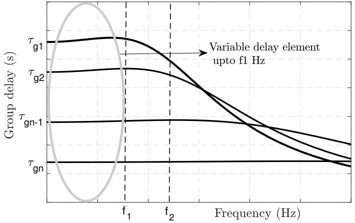
Variable-Delay APF*





- Changing L,C values varies delay τ_g
- In Gm-C implementation of APF
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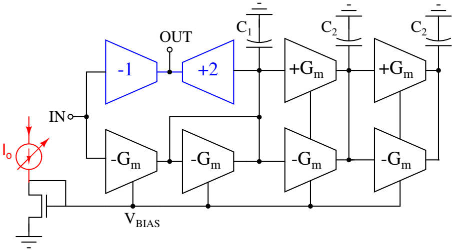
- Delay tuning by changing the Gm of the filter and no need of high-Q varactors



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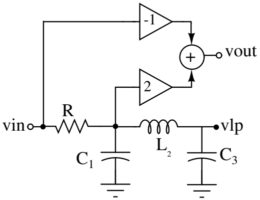
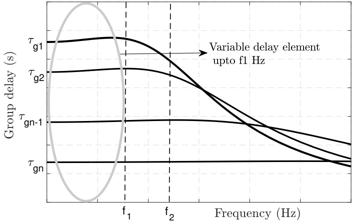
Variable-Delay APF*





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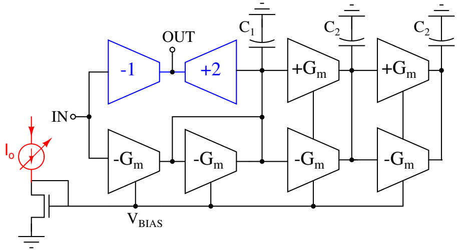
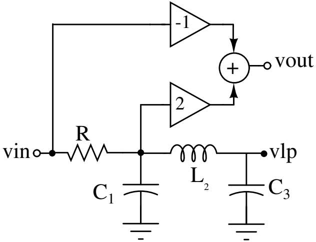


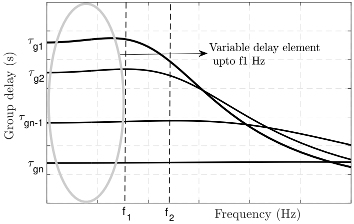
- Programmable delay APFs in TTD and Pulse expansion (compression) techniques.

* Mayur S.M. et al. IEEE ISCAS 2022

Overview

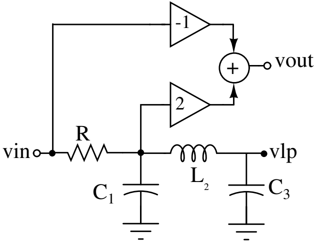
Variable-Delay All-Pass Filter (APF) as TTD *

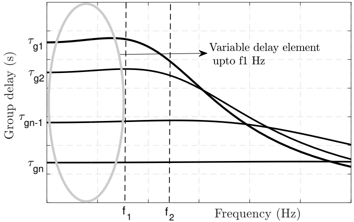




- Varying L,C changes the delay τ_g

Variable-Delay All-Pass Filter (APF) as TTD *





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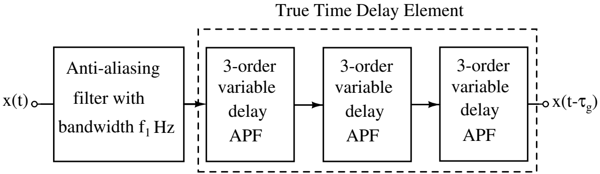
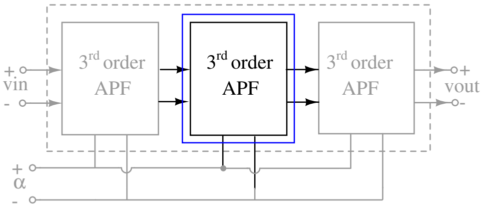
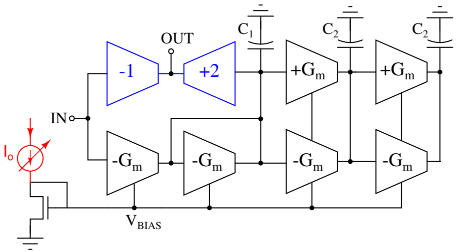


* Mayur S M et al. in IEEE ISCAS-2022.

Delay Tuning Technique

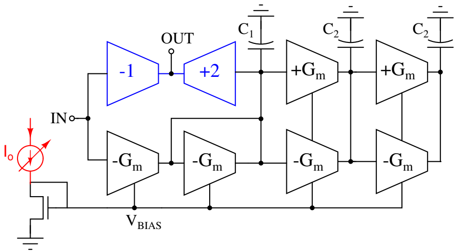


Delay Tuning Technique



- In Gm-C implementation of APF, Gm of the filter is varied for delay tuning

Delay Tuning Technique



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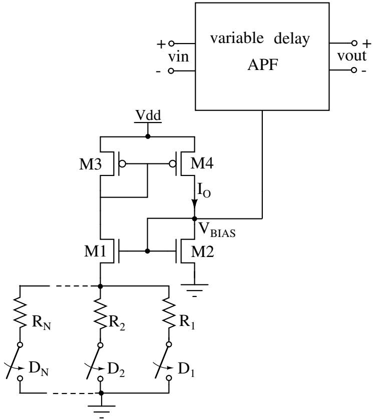
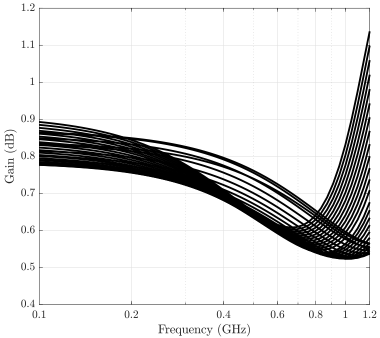
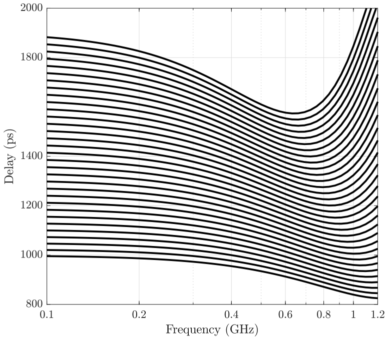


- Constant Gm biasing with variable current is used for delay tuning

Simulation Results

- Implementation in TSMC 65 nm technology with supply voltage of 1.2 V
- Delay varies from 1 ns to 1.8 ns with gain deviation of ±0.16 dB





Comparision with TTD architectures

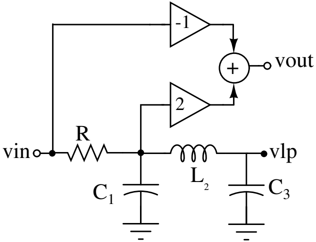
Parameters	This work	This work	[MTTS-2015]	[JSSC-2015]	[JSSC-2017]	[MWCL-2021]
Technology (nm)	65	65	130	140	130	65
Architecture	Gm-C	LPF with correc.	Passive Trombone	Gm-C	Gm-C	Gm-RC
Maximum delay(ps)	1800	1150	400	550	1700	55
Bandwidth (GHz)	DC-1	DC-1	1-20	1-2.5	DC-2	0.4-2
Delay Range (ps)	800	400	350	550	1450	27
Delay Variation (ps)	$\pm$120	$\pm$5	$\pm$8	$\pm$10	$\pm$140	$\pm$2
Gain Variation (dB)	$\pm$0.16	$\pm$2.5	$\pm$15	$\pm$1.4	$\pm$0.7	$\pm$0.375
Power (mW)	32	36	6	450	364	2.47
FOM-1	-27.95	-20.92	-60.89	-5.26	-18.02	-24.85
FOM-2	-38.09	-47.95	-57.69	-31.25	-30.91	-40.69

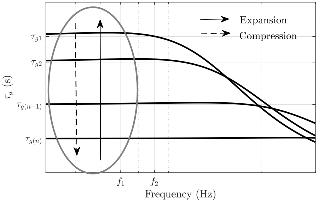
$$\text{FOM-1}=20\log(\text{Power}/(\text{delay range} \times \text{bandwidth}))$$

$$\text{FOM-2}=\text{FOM-1}+20\log\left(\frac{2\times\text{delay variation}}{\text{delay range}}\right)+2\times\text{gain variation}$$

Overview

Variable-Delay APF for Expansion (Compression)





- Varying L and C changes delay $\tau_{g(n)}$

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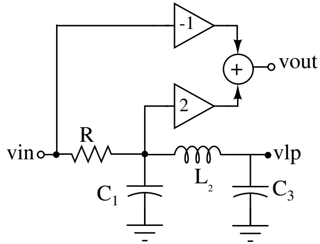

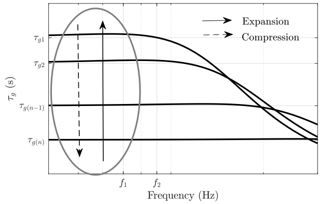

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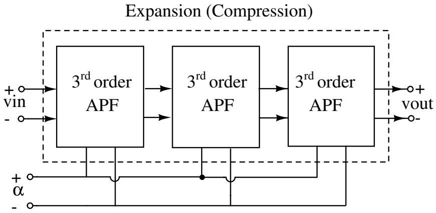
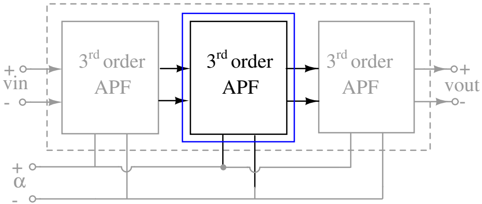


Delay Tuning Technique



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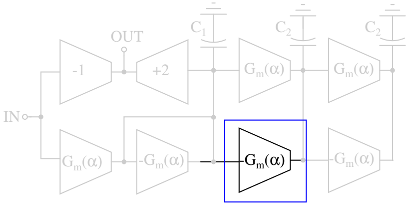

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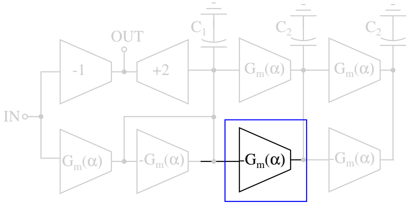

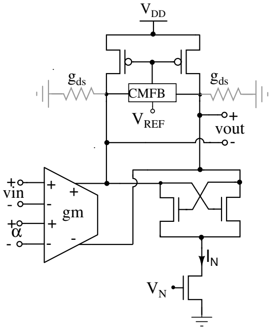

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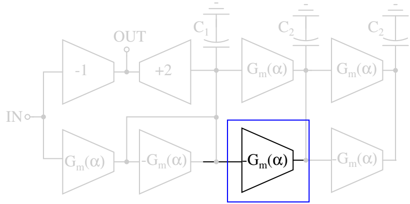
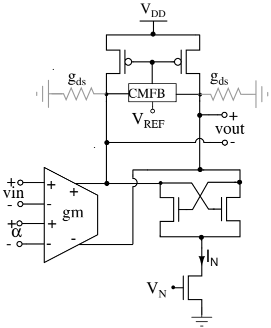
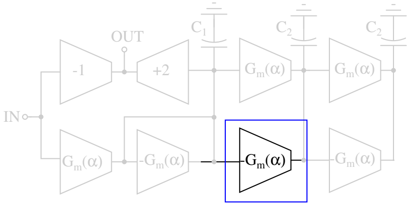
Delay Tuning Technique

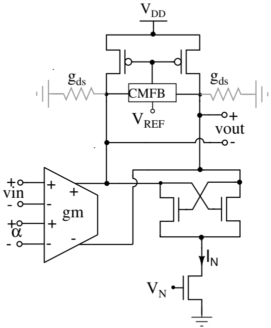




- In Gm-C implementation of APF, Gm of the filter is varied for delay tuning
- Programmable Gm realized from Gilbert multiplier for expansion and compression

Delay Tuning Technique





- In Gm-C implementation of APF, Gm of the filter is varied for delay tuning

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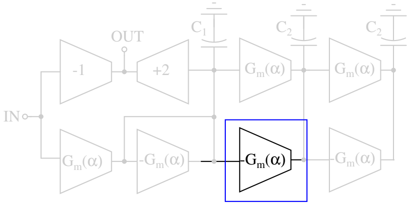
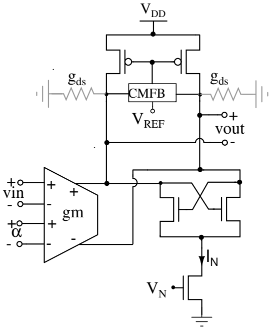
- Decreasing α : Expansion

Delay Tuning Technique





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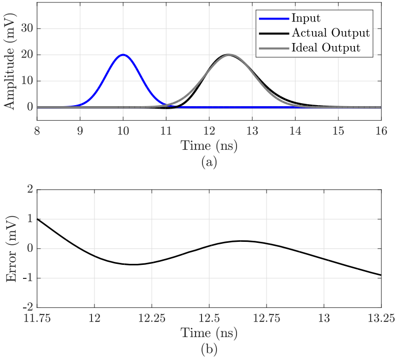
- Programmable Gm realized from Gilbert multiplier for expansion and compression
- Decreasing α : Expansion
- Increasing α : Compression

Simulation Results

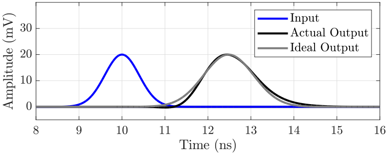
Simulated in TSMC 65-nm technology



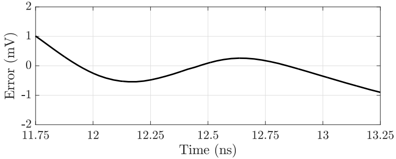
- Ideal Gaussian input pulse 0.944 ns
- Stretch factor 1.48

Simulation Results

Simulated in TSMC 65-nm technology

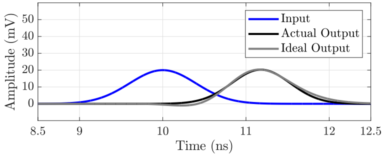


(a)



(b)

- Ideal Gaussian input pulse 0.944 ns
- Stretch factor 1.48



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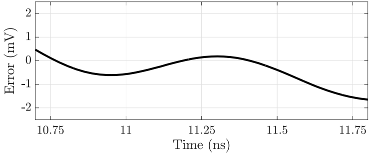


(b)

- Ideal Gaussian input pulse 0.944 ns
- Compression factor 0.825

Comparison between APF and LPF based Expansion Techniques

- Expansion with 1-stage variable-delay APF
- Expansion with 3-stage variable-delay LPF
- Summary of schematic simulations at 65-nm technology

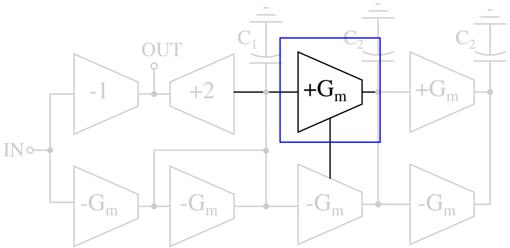
Technique	APF	LPF
Pulse width (ns)	0.944	0.944
Stretch factor	1.04	1.04
RMSE (mV)	0.318	0.533
Power dissipation (mW)	7.38	6.78
FOM(mW/mV)	23.21	12.72

FOM=Power/RMSE

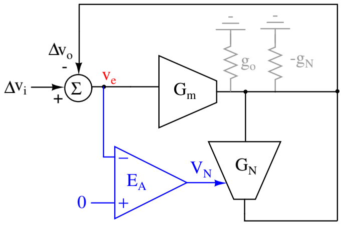
- Trade-off with power and accuracy

Overview

Inline Calibration for High Gain OTA

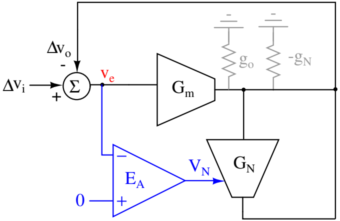


- Identify the steady state error v_e
- Force this error v_e to 0 by another negative feedback.



Principle of Inline Calibration

- Identify the steady state error v_e
- Force this error v_e to 0 by another negative feedback.



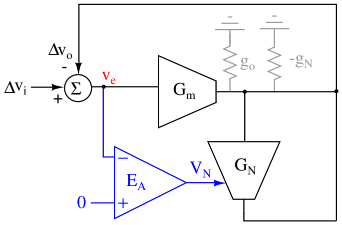
- Effective output conductance

$$g_{oN} = g_o - g_N$$

- DC Gain of OTA

$$A_{oN} = G_m / g_{oN}$$

- Identify the steady state error v_e
- Force this error v_e to 0 by another negative feedback.



- Effective output conductance

$$g_{oN} = g_o - g_N$$

- DC Gain of OTA

$$A_{oN} = G_m / g_{oN}$$

- Steady state error

$$v_e = \Delta v_i \; g_{oN} / (G_m + g_{oN})$$

Principle of Inline Calibration

- Identify the steady state error v_e
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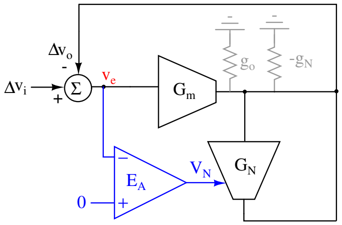

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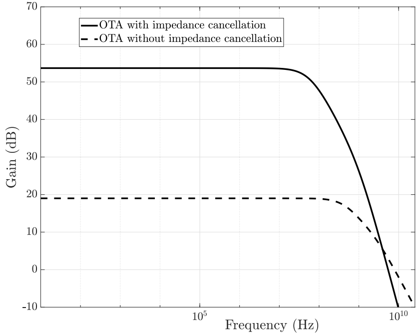
$$A_{oN} = G_m / g_{oN}$$

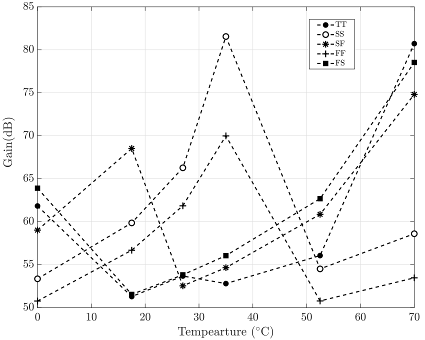
- Steady state error

$$v_e = \Delta v_i \; g_{oN} / (G_m + g_{oN})$$

- Negative feedback ensures $v_e = 0$ and hence $g_o = g_N$

Simulation Results *



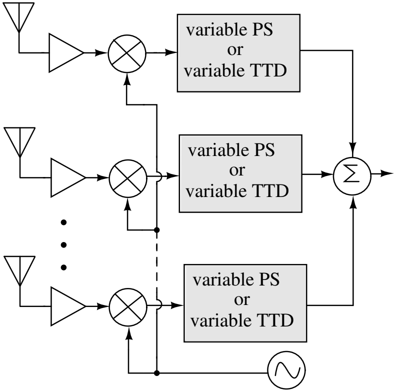


- Proposed OTA is implemented in TSMC 65-nm technology
- Gain of OTA is increased from 18 dB to 55 dB
- Minimum of 50 dB maintained across all the process corners

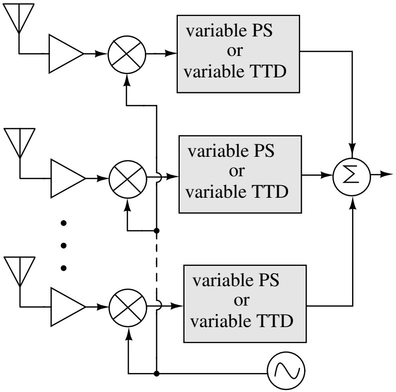
* presented in APSCCAS 2023

Overview

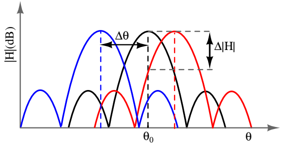
Effect of TTD Non-Idealities



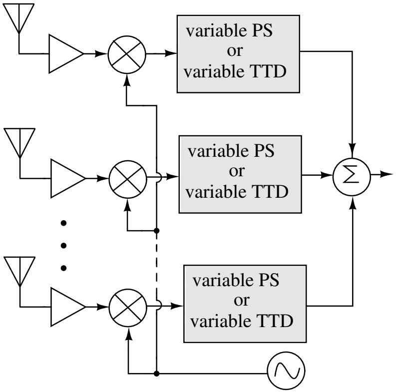
Effect of TTD Non-Idealities



- Phase Shifters suffer from Beam-squints for baseband with wide bandwidth



Effect of TTD Non-Idealities



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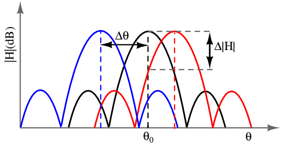
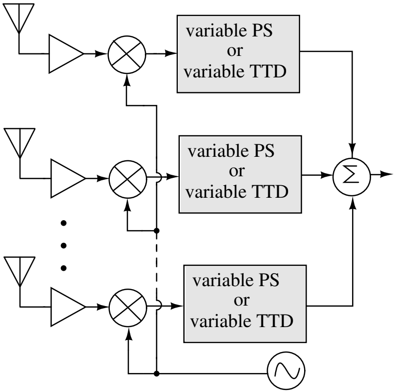


- TTD circuit for beamforming

Effect of TTD Non-Idealities



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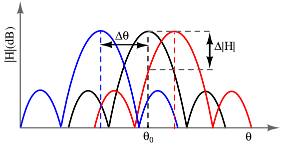

- TTD circuit for beamforming
- Ideal TTDs must maintain flat delay and constant gain

Performance summary of various reported TTDs

Performance summary of different LC based TTDs

Process-Technology	Topology	Bandwidth (GHz)	Delay resolution (ps)	Maximum delay (ps)	Delay variation(%)	Gain variation(dB)
CMOS 90-nm	Tuned LC	DC-8	14	40	10	±0.5
CMOS 45-nm	Tuned LC	11-24	10	50	11	–
GaAs 250-nm	Switched Line	6-18	1	255	2.43	±2
CMOS 130-nm	Switched Line	8-16	1.56	198.4	2.16	±1
SiGe BiCMOS 180-nm	Switched Line	25-50	4	124	1.6	±4.4
CMOS 130-nm	Trombone	1-15	15	225	14	±3
SOI 45-nm	Trombone	135-145	0.3	11.57	1.4	±0.57

Performance summary of different APF based TTDs

Process-Technology	Topology	Bandwidth (GHz)	Delay resolution (ps)	Maximum delay (ps)	Delay variation(%)	Gain variation(dB)
CMOS 130-nm	Passive-Trombone	1-20	5.6	400	17.8	±15
CMOS 180-nm	Passive-Trombone	8-18	15.6	109.3	19	±1
CMOS 28-nm	Passive-Switched Line	3-30	4.6	68.5	4.3	±1.6
CMOS 180-nm	Passive-Internal switched	8-18	3.9	125	21	±2
CMOS 130-nm	Active Gm-C	DC-2	250	1700	24	±0.7
CMOS 140-nm	Active Gm-C	1-2.5	13	550	1.8	±1.4
CMOS 65-nm	Active Gm-RC	0.4-2.5	2	50	2.49	±0.375

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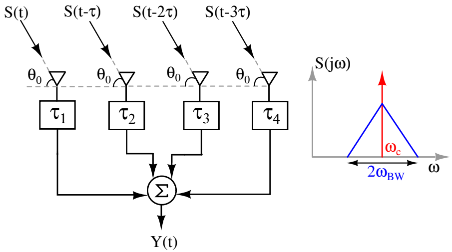
RF Beamforming with non-ideal TTDs



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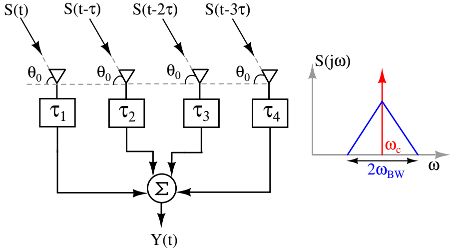

- 5G mm-wave receiver with f_c=28 GHz and f_{BW}=1 GHz

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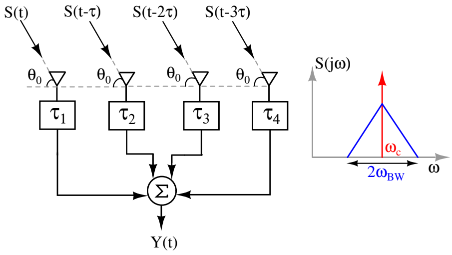

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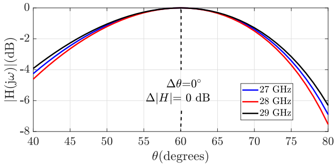

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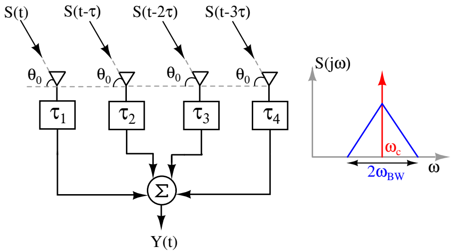

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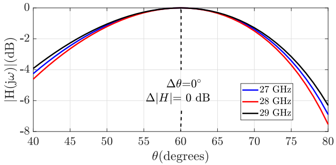
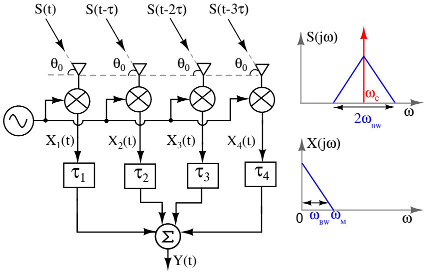


Δ|H| and $\Delta\theta$ observed for different levels of delay and gain variations

Delay variation(%)	Gain variation (%)	Δ|H|(dB)	$\Delta\theta^\circ$
10	–	-0.144	3
20	–	-0.583	6
–	10	-0.915	0
–	20	-1.93	0
10	10	-1.059	3
20	20	-2.521	6

BB Beamforming with non-ideal TTDs



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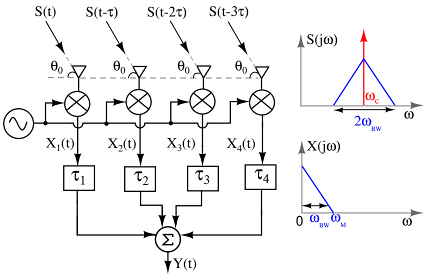

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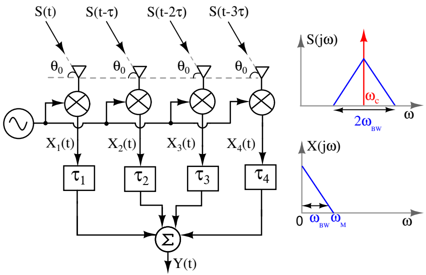

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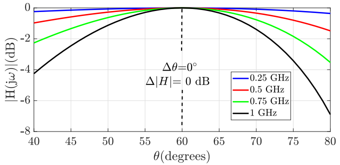

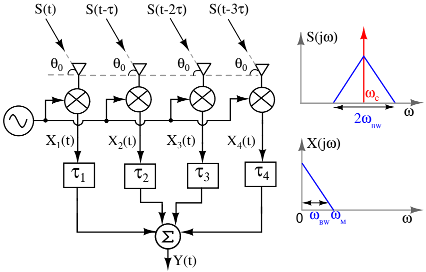

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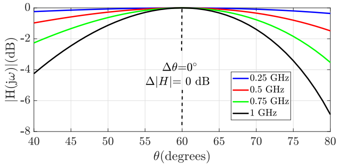

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Delay variation(%)	Gain variation (%)	Δ|H|(dB)	$\Delta\theta^\circ$
10	–	-0.134	3
20	–	-0.543	6
–	10	-0.915	0
–	20	-1.938	0
10	10	-1.059	3
20	20	-2.481	6

Comparision of Practical PS and Practical TTDs

PS	Phase shift variation(%)	Gain variation(%)	$\Delta\|H\|$(dB)	$\Delta\theta^\circ$
	10	10	-1.256	9
	20	20	-2.825	15
TTD	Delay variation(%)	Gain variation (%)	$\Delta\|H\|$(dB)	$\Delta\theta^\circ$
	10	10	-1.059	3
	20	20	-2.521	6

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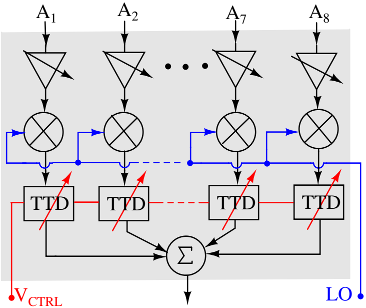
Immediate future work

- Respin tapeout of TTD and expansion designs fabricated in 65 nm technology
- Complete 1×8 receiver architecture with proposed LPF based TTD circuits



- Use of inline calibration technique for OTAs in Gm-C APF filter design

Future work

- Optimization of LPF and APF designs considering distortion and noise
- Delay tuning of complex filters with any impedance Z=R $\pm$ j X using the Miller effect
- A variable order LPF with controlled delays using the principle of the Miller effect
- Tunable band-pass filters realized from variable-delay LPFs which are used in GNS/GPS receivers
- Proposed delay tuning techniques in the application of equalizers

Patent and Publications from This Work

1. Naveen K, Mayur S, “A Circuit for Expanding, Compressing, or Delaying a Signal”, Indian Patent 555660.
2. Mayur S.M, Ganga K. M.,and Naveen K,“A True Time Delay Element using Cascaded Variable Bandwidth All Pass Filters”, in *2022 IEEE International Symposium of Circuits and Systems (ISCAS).*
3. Mayur S.M and Naveen K, “Mismatch Tolerant Negative Conductance Load Tuning for High Gain OTAs”,in *2023 IEEE Asia Pacific Conference on Circuits and Systems (APCCAS).*
4. Mayur S.M and Naveen K.“A Review of True Time Delay based Beamforming and Analysis of some Practical Implications”, accepted in *IEEE Microwave Magazine.*

THANK YOU